AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 17 · STUDENT LESSON

Precision and Recall

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the positive. Name the counted group. Keep the denominator.

Explain precision and recall as distinct questions and make threshold choices under declared priorities.

Keep one habit: Complete worked matrix → pictured denominator contrast → zero-denominator cases → complete threshold/cost lab → independent19-record matrix.

MAIN LESSON · SECTION 1

Which group belongs under this fraction?

> Precision counts correct flags among flags; recall counts found positives among reference positives.

Learning goals

I can choose the **denominator** that answers the question.

I can calculate **precision** and **recall** with a declared positive class.

I can report **undefined** ratios honestly.

Recall imbalance and hidden misses

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.
- Why can 80 percent accuracy hide **miss**ed blocked cases?

End goal

By the end, I can explain **precision** and **recall** using a traced example and a stated limitation.

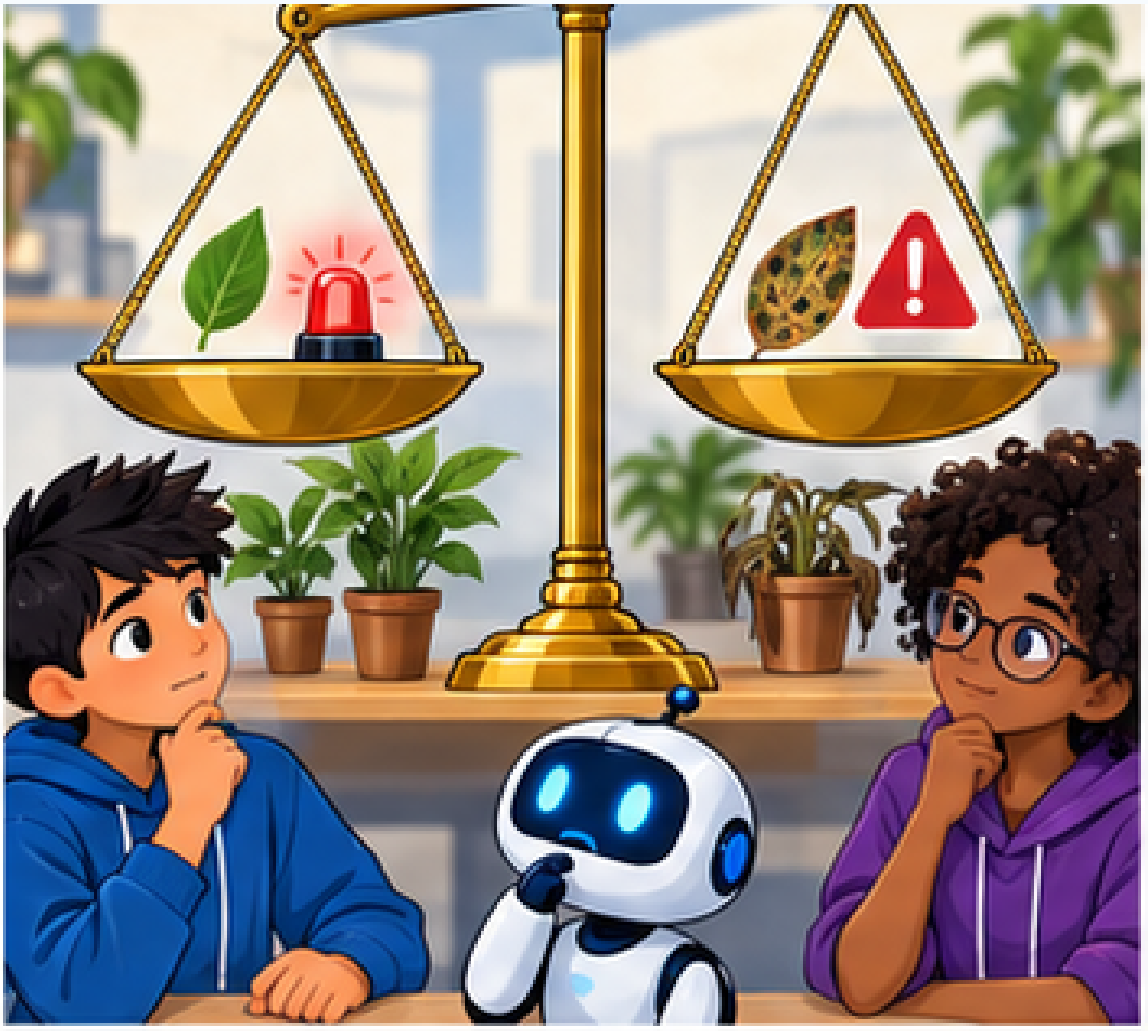
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Precision measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.



LEARNING TARGET TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Ask two different questions about the same true positives

> A fraction needs a named positive class and labelled axes.

Blocked-positive definitions and source scope

Blocked remains positive. **Precision** asks: of the cases flagged blocked, how many were actually blocked according to the reference? **Recall** asks: of all reference-blocked cases, how many were found?

Both use true positives, but they answer different questions. Keep reference rows and **prediction** columns ordered clear then blocked.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

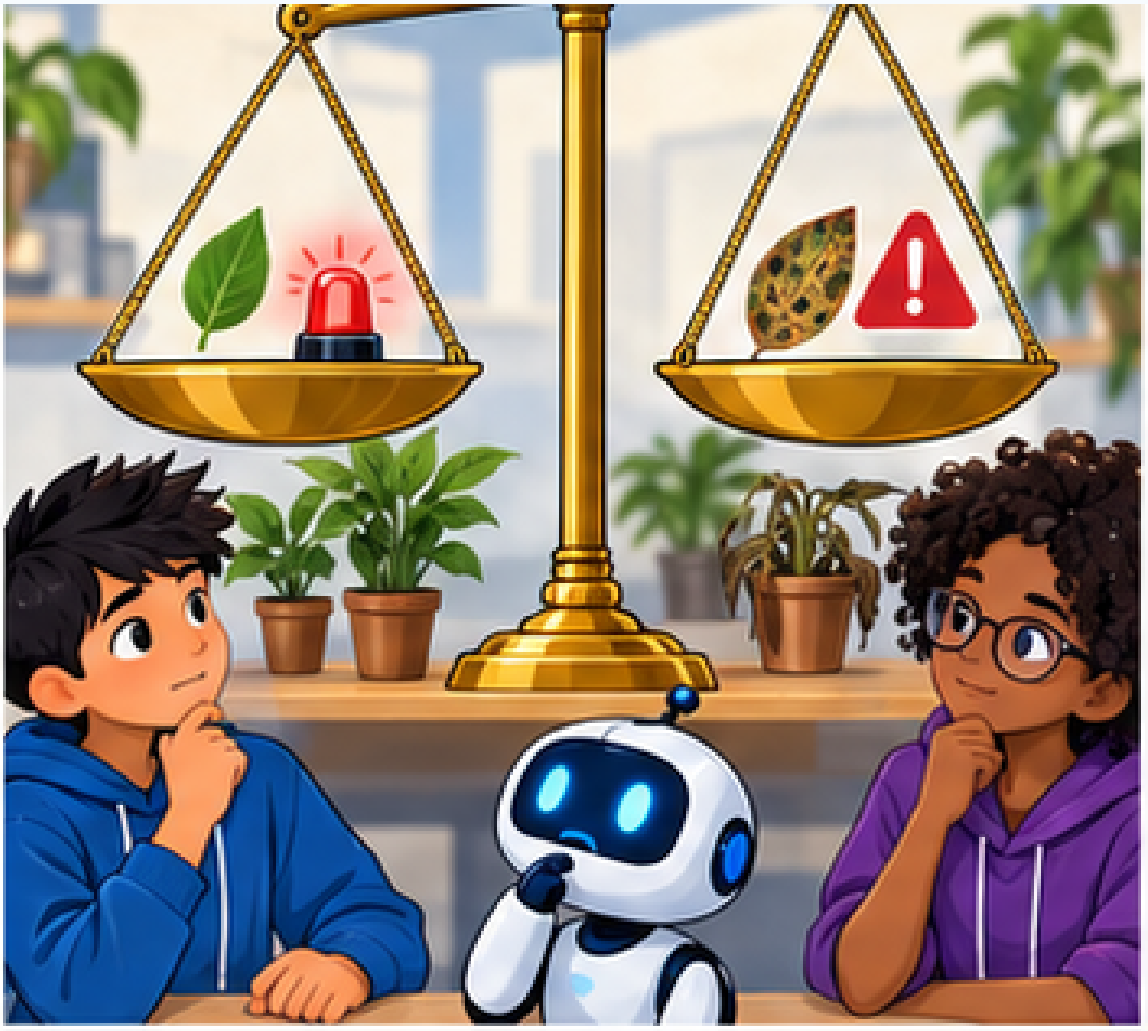
Earlier weeks supplied the model-building **process**; this week examines **precision recall** and **denominator**s.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **precision** and **recall** to solve a model-building problem. What information do they have? What important **evidence** is still **miss**ing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Trace TP through two different denominators

> Keep ordinary accuracy as a third, separate fraction.

Complete worked matrix and denominator counts

For matrix [[12, 4],[1, 3]], TN=12, FP=4, FN=1, TP=3. There are seven predicted positives and four reference positives. **Precision** is 3/7, about 42.9 percent. **Recall** is 3/4=75 percent. Accuracy is 15/20=75 percent.

A correct **denominator** matters more than memorizing a percentage.

Optional complete guarded-ratio script

```
tn, fp, fn, tp = 12, 4, 1, 3
precision = tp / (tp + fp) if tp + fp else None
recall = tp / (tp + fn) if tp + fn else None
accuracy = (tn + tp) / (tn + fp + fn + tp)
print(precision, recall, accuracy)
# None means unavailable, not zero or perfect.
```

Predict before running

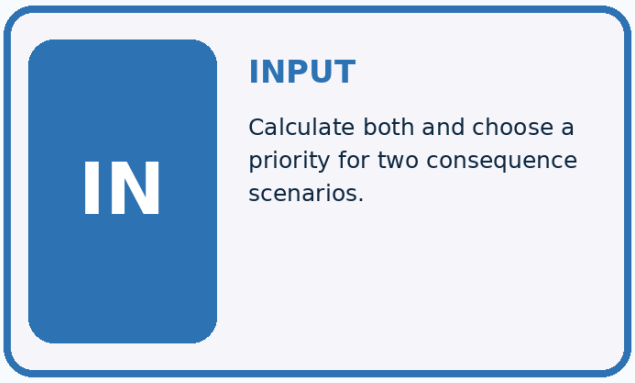
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

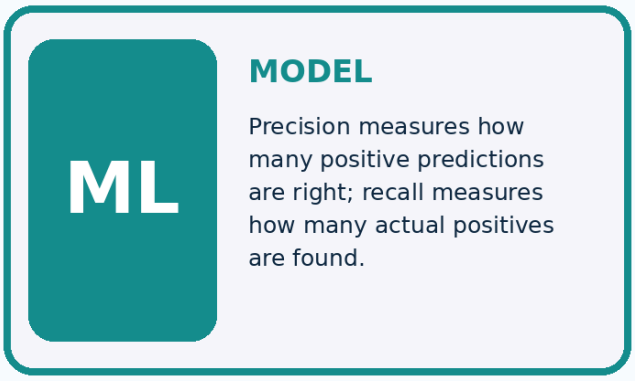
▼ **Earlier wording, alternatives and picture transcripts**

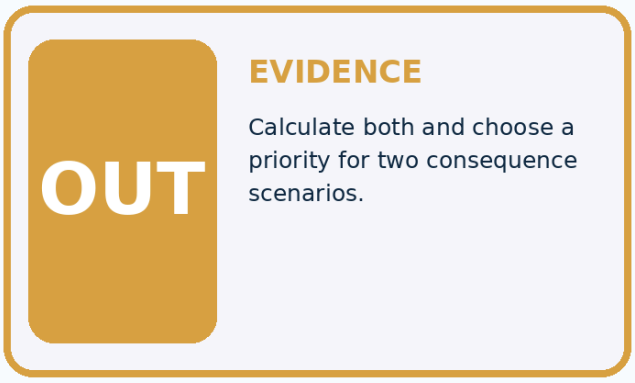
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Calculate both and choose a priority for two consequence scenarios.

MODEL IDEA **Precision** measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Confusion counts: TP=8, FP=2, FN=4, TN=16. **Precision** = 8 ÷ (8+2) = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

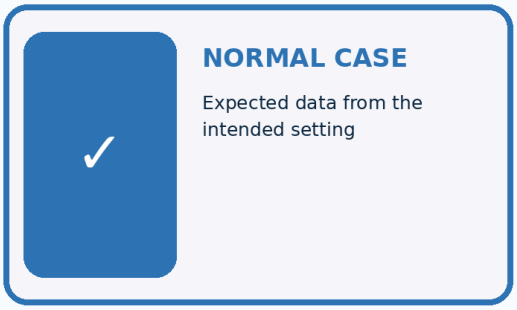
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

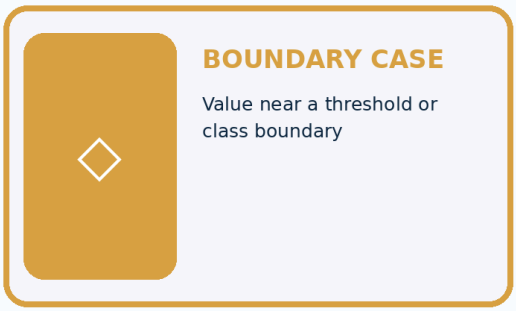
Compare reliable flags with finding the positives

> A metric name alone does not say which errors matter most.

Four evaluation conditions



NORMAL Expected data and expected behavior.

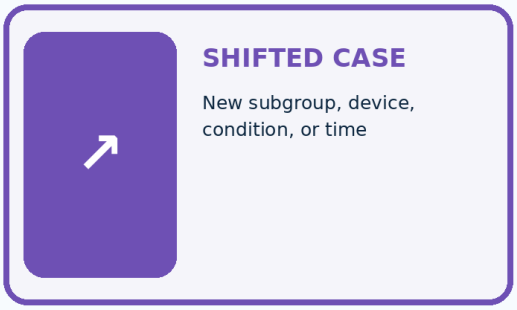


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Correct two false negatives into true positives.

▼ **Check after predicting**

Prediction: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

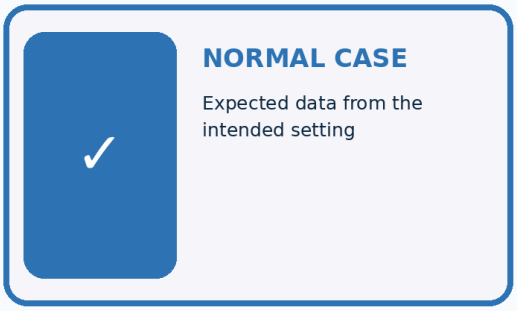
Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

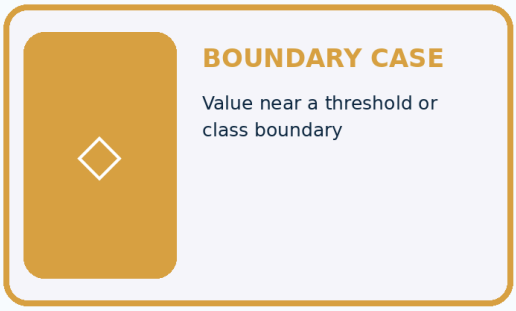
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

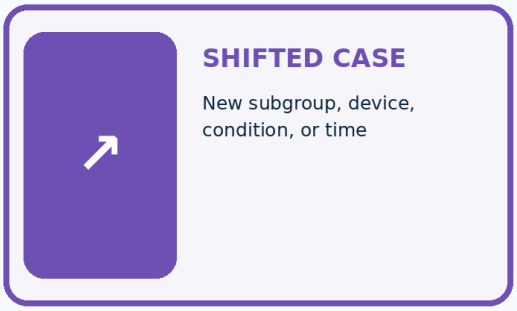
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Put the two counted groups beside the picture

> The same numerator does not make two ratios interchangeable.

Complete pictured counts and separate-fixture scope

Put **precision** and **recall denominator**s on the picture instead of memorising names.

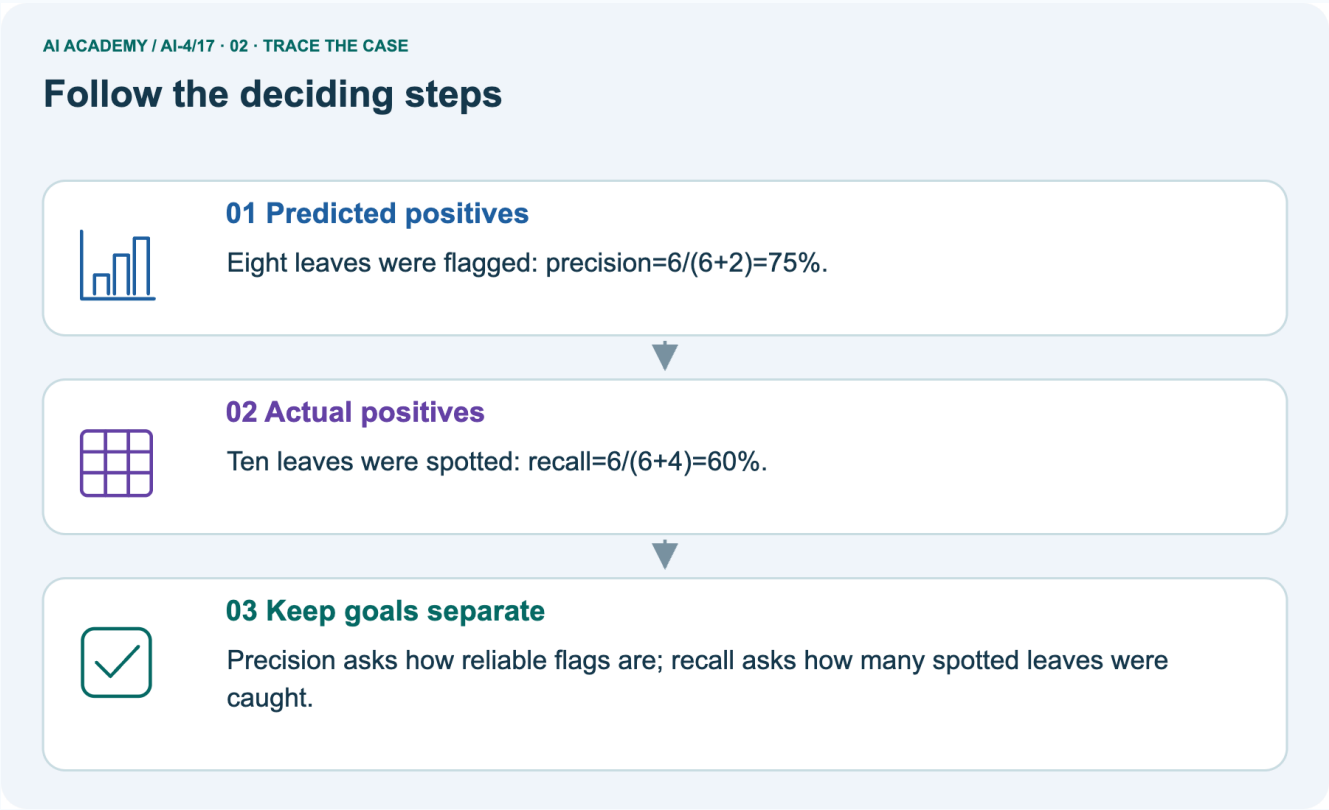
Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main corridor matrix is [[12,4],[1,3]] with clear/blocked axes: TP3,FP4,FN1,TN12;precision3/7,recall3/4,accuracy15/20. Picture instead uses spotted positive and TP6,FP2,FN4,TN8:precision6/8=75%,recall6/10=60%. Moving two pictured FN decisions to TP gives TP8,FN2,FP2,TN8:precision8/10,recall8/10. These denominators happen to become equal; they answer different questions.

If the move is a human-reviewed correction, preserve the original model-only predictions and report reviewed outcomes separately.

Trace flags, actual positives and their ratios



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

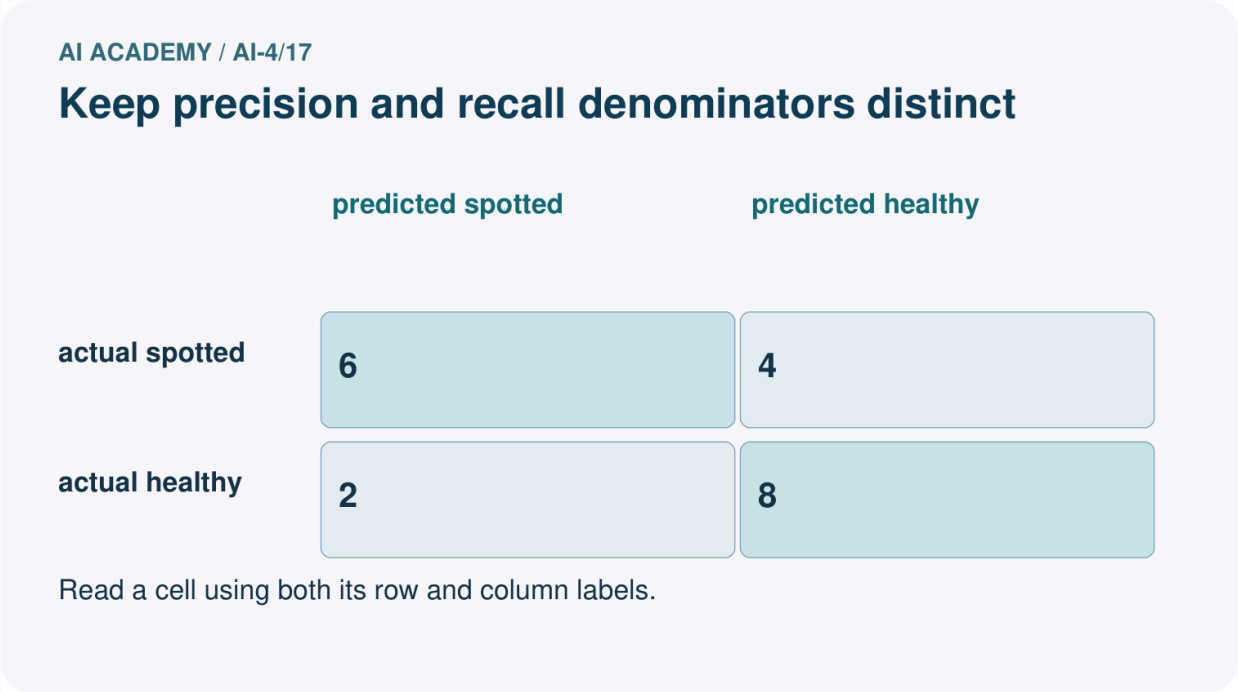
STEP	WHAT HAPPENS AND WHY
1. Predicted positives	Eight leaves were flagged: precision=6/(6+2)=75%.
2. Actual positives	Ten leaves were spotted: recall=6/(6+4)=60%.
3. Keep goals separate	Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original denominator picture

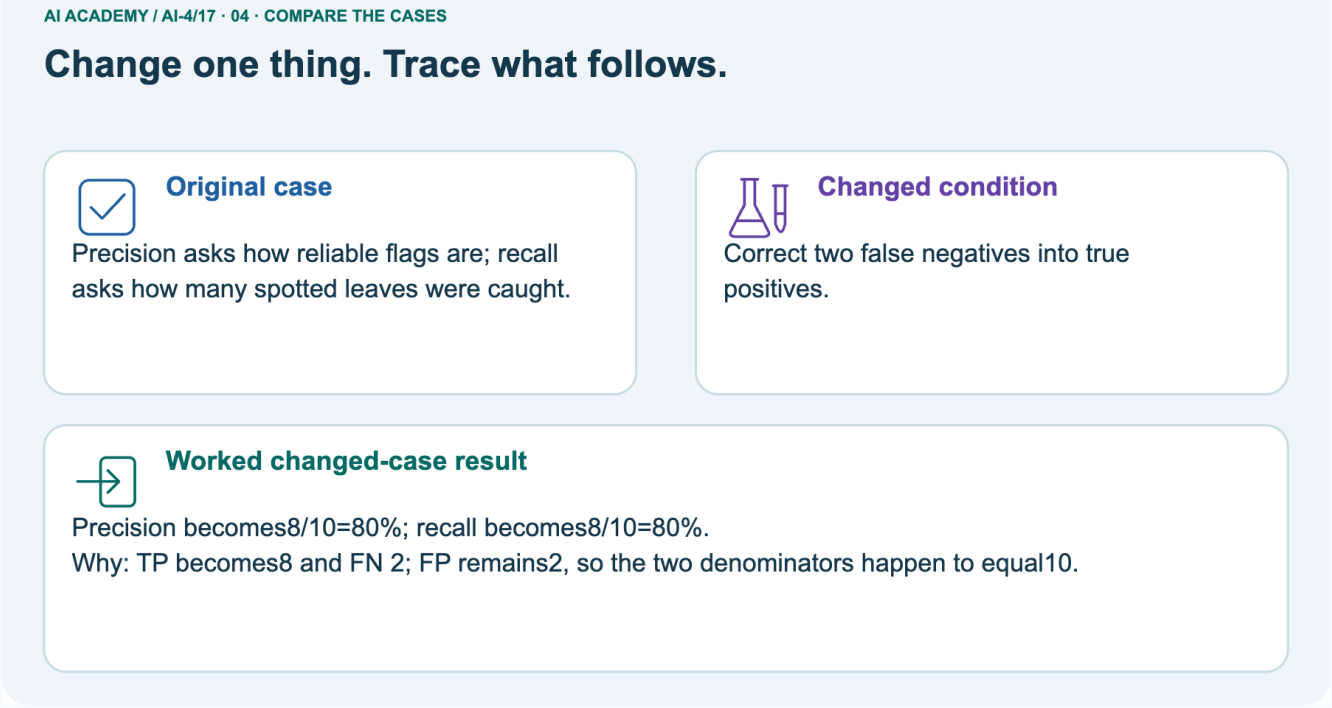


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict two false negatives moving to true positives

- Change one thing: Correct two false negatives into true positives. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain zero-denominator and ratio limits

- If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

- Read the labelled picture: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the corridor ratios

Precision recall and denominators at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Precision	3 correct flags	7 total flags
Recall	3 found blocked cases	4 reference blocked
Accuracy	15 correct labels	20 total records

Draw a box around the predicted-blocked column for precision and around the reference-blocked row for recall. Their overlap is TP. A model can flag nearly everything and find most blocked cases while producing many false alarms. Conversely, a model can flag very few cases accurately while missing many positives.

Calculate precision, recall and accuracy for the worked matrix. Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 17 • WORKED EVIDENCE

1. Confusion counts: TP=8, FP=2, FN=4, TN=16.
2. Precision = 8 ÷ (8+2) = 80%.
3. Recall = 8 ÷ (8+4) ≈ 67%.
4. Choose threshold and human review by comparing the cost of two false alarms with four misses.

STEP 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.

STEP 2 **Precision** = 8 ÷ (8+2) = **80%**.

STEP 3 **Recall** = 8 ÷ (8+4) ≈ 67%.

STEP 4 Choose threshold and human review by comparing the cost of two **false alarm**s with four **miss**es.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

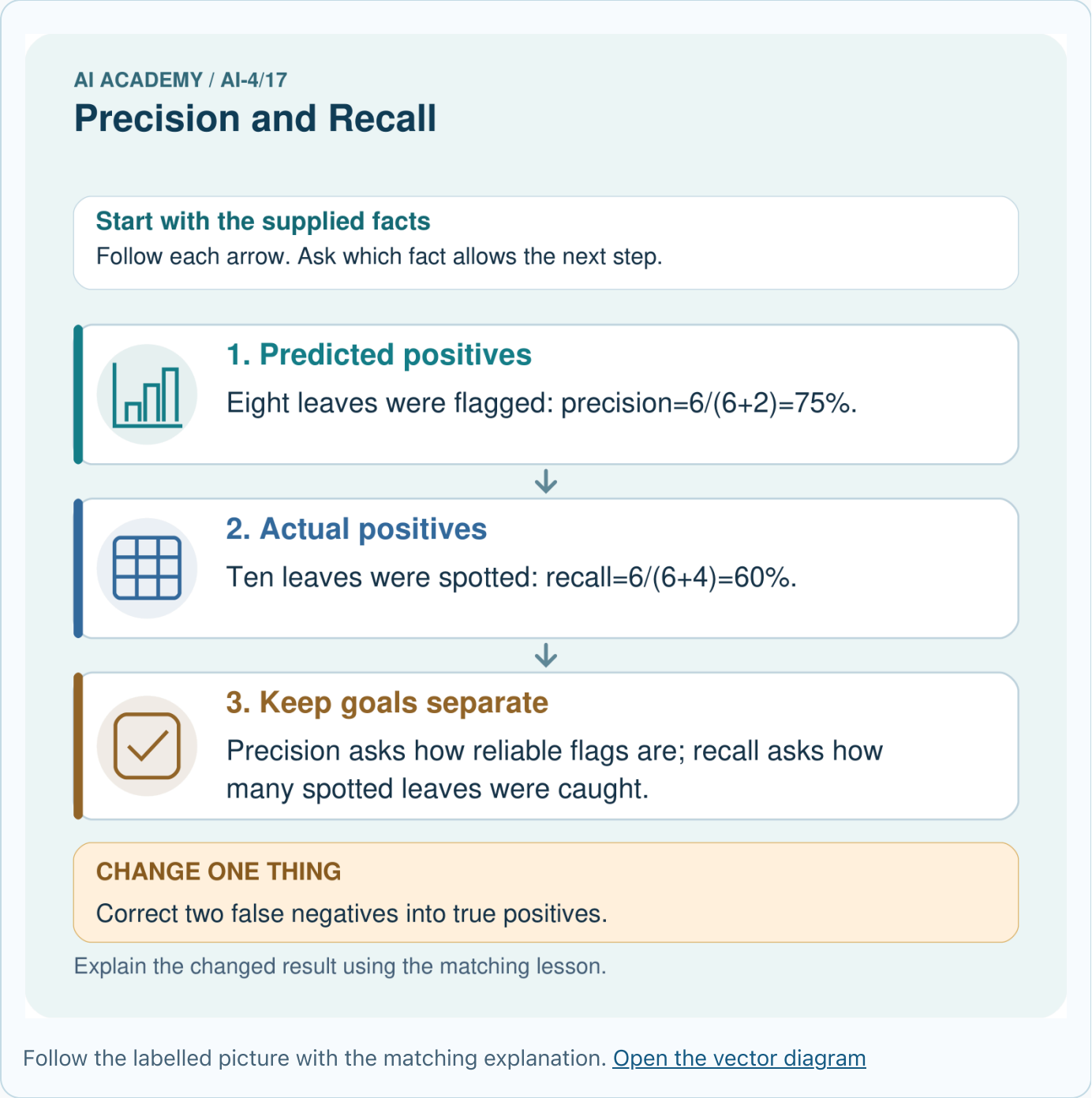
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Precision and Recall

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the wrong denominator before fixing a percentage

> Unavailable evidence is different from zero success or perfect success.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE **Precision** and **recall** are two names for the same accuracy calculation.

Explain the method and limitations

- Ratios describe this evaluation set, not a guarantee for an individual case. Library zero-division policies differ; document yours. Do not select a metric only after seeing which one makes a model look best.

> **A team reports recall 100 percent but hides precision and counts. What is missing?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Precision and recall are two names for the same accuracy calculation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise each ratio and its absent-denominator case

> Record counts so an undefined metric stays interpretable.

Try W2 with the complete matrix

Use workbook W2 for the connected example “**Precision recall** and **denominator**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Compare no predicted positives and no reference positives

With [[16, 0],[4, 0]], there are no predicted positives. **Precision** is **undefined** because its **denominator** is zero; **recall** is 0/4=0. With [[16, 4],[0, 0]], **recall** is **undefined** because no reference positives were evaluated.

Record counts and the **undefined** result instead of calling it perfect.

Explain the source of each calculation

- Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the original pictured denominators

- Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the new independent matrix

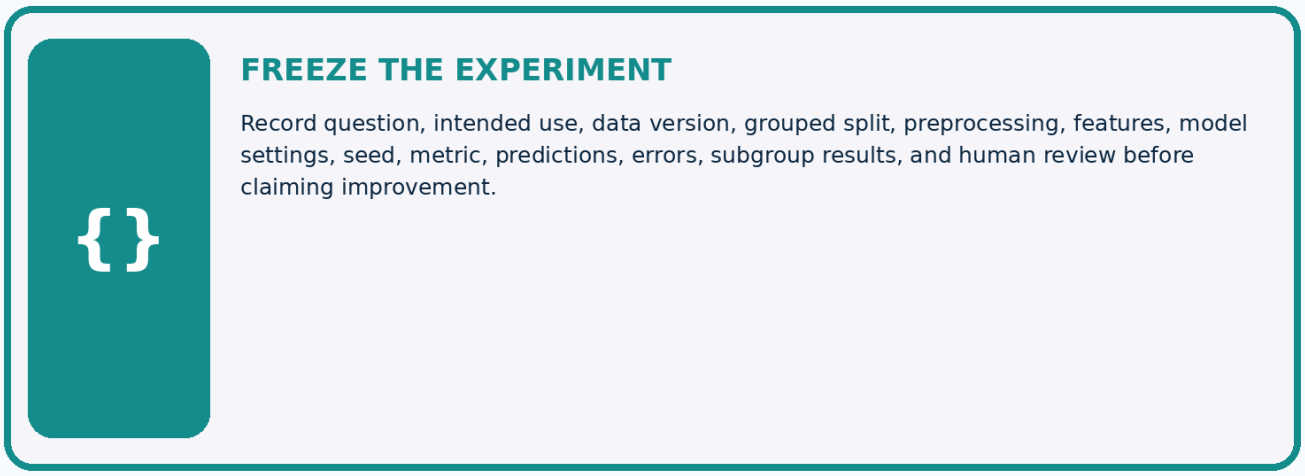
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + **denominator**s

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Calculate both ratios

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete two-threshold metric and consequence lab

> Every score, reference and policy value needed for the comparison is supplied.

Complete ten fictional score/outcome records

Case ID	Supplied score for positive	Observed positive?
T01	0.9	1
T02	0.8	1
T03	0.7	1
T04	0.6	0
T05	0.5	1
T06	0.4	0
T07	0.3	1
T08	0.2	0
T09	0.1	0

Case ID	Supplied score for positive	Observed positive?
T10	0.0	0

Threshold contract and counted errors

Metric Tradeoff Lab uses these ten fictional development records, with1positive/0negative and prediction positive when score>=threshold. Scores are supplied rankings for this exercise; calibration is not established. Predeclare thresholds0.4and0.7. At0.4:TP4,FP2,FN1,TN3;precision4/6,recall4/5,accuracy7/10. At0.7:TP3,FP0,FN2,TN5;precision3/3,recall3/5,accuracy8/10. T03at0.7is included by equality.

These threshold comparisons use the same references and records; a final independent evaluation is not supplied.

Two explicit fictional priorities

Fictional consequence policy	Threshold0.4	Threshold0.7
A: each miss5points,each false alarm1point	FN1×5+FP2×1=7points	FN2×5+FP0=10points
B: each miss1point,each false alarm5points	FN1+FP2×5=11points	FN2+FP0=2points
Interpretation	Policy A prefers0.4on this fixture.	Policy B prefers0.7on this fixture; these invented costs do not establish real-world harms.

Normal, equality, no-flags and no-reference-positive probes

Complete denominator probe	Expected bounded check
Normal ten-record threshold0.4	Precision4/6,recall4/5,accuracy7/10.
Equality threshold0.7onT03score0.7/positive	Positive prediction; keep>=distinct from>.

Complete denominator probe	Expected bounded check
Threshold1.0on these same ten rows	No positive predictions:precision unavailable,recall0/5=0.
Separate no-reference-positive fixture: U1score0.8/outcome0,U2score0.2/outcome0,threshold0.7	Precision0/1=0;recall unavailable. No evidence of positive detection.

Predict, count, calculate and defend

1. Predict both ten-row label lists before calculating TP/FP/FN/TN. Show each numerator and denominator, then use the two declared fictional policies to defend a conditional choice. Keep threshold equality, absent denominators and score meaning explicit. Higher thresholds do not guarantee better precision on every possible dataset; this fixture supplies one observed tradeoff.

   Preserve the complete development comparison and reserve later independent data for a final evaluation. Keep the selected19-record matrix for an unaided first attempt.

Optional complete threshold program

```
rows = [('T01', .9, 1), ('T02', .8, 1), ('T03', .7, 1),
        ('T04', .6, 0), ('T05', .5, 1), ('T06', .4, 0),
        ('T07', .3, 1), ('T08', .2, 0), ('T09', .1, 0), ('T10', .0, 0)]


def evaluate(rows, threshold):
    if not 0 <= threshold <= 1:
        raise ValueError('Threshold outside declared 0..1 range')
    if len({i for i, score, actual in rows}) != len(rows):
        raise ValueError('Duplicate case ID')
    tp = fp = fn = tn = 0
    for item_id, score, actual in rows:
        if not 0 <= score <= 1 or actual not in (0, 1):
            raise ValueError('Need declared-range score and observed binary outcome')
        prediction = int(score >= threshold)
        tp += int(actual == 1 and prediction == 1)
        fp += int(actual == 0 and prediction == 1)
        fn += int(actual == 1 and prediction == 0)
        tn += int(actual == 0 and prediction == 0)
```

```
    precision = tp / (tp + fp) if tp + fp else None
    recall = tp / (tp + fn) if tp + fn else None
    accuracy = (tp + tn) / len(rows) if rows else None
    return {'tp': tp, 'fp': fp, 'fn': fn, 'tn': tn,
            'precision': precision, 'recall': recall, 'accuracy': accuracy}


for threshold in [.4, .7]:
    print(threshold, evaluate(rows, threshold))
# These are development comparisons, not untouched final-test conclusions.
```

Retain evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

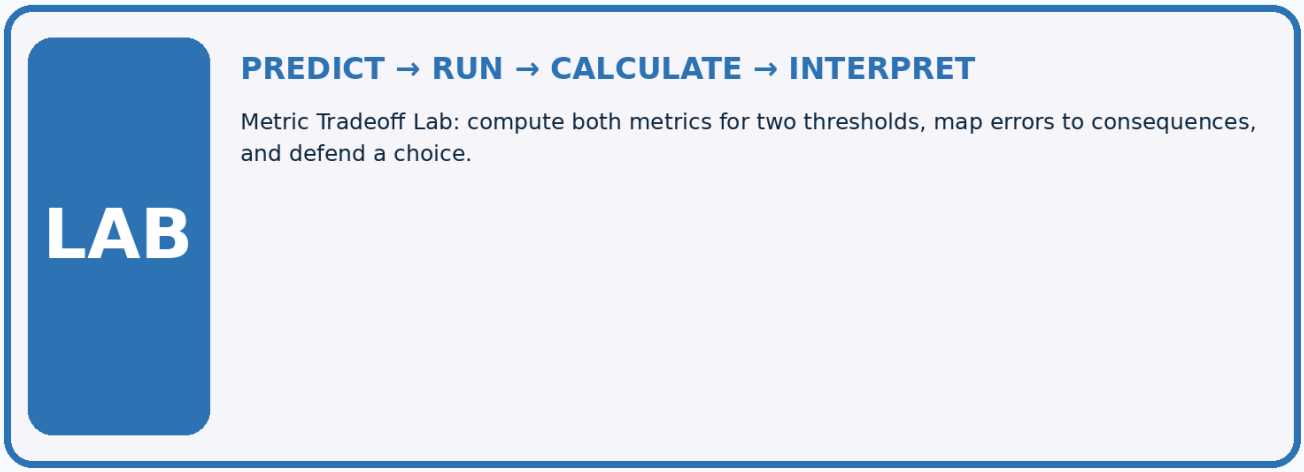
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.

3. Run one normal, one boundary, one **miss**ing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: interpret three ratios from a new matrix

> Name the positive class and each counted group.

Complete independent matrix and axis scope

Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

Use clear/blocked on both axes, reference rows and prediction columns, with blocked positive. This supplied matrix has19records; do not use the20-case worked or pictured counts. State predicted-positive and reference-positive totals before dividing. No row IDs, score list or alternative threshold is supplied, so do not invent a threshold comparison for this independent task.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or **miss**ing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project threshold and its intended priority together

> A choice needs an error consequence as well as a ratio.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

Save the positive definition, score meaning, exact threshold/equality policy, dataset version and reference/prediction pairs. Report precision, recall, accuracy, their denominators and unavailable results. Record the development consequence policy and human reviewer separately.

Threshold selection on development evidence is not an untouched final-test result; a review threshold must be justified for the intended use rather than inferred from one attractive metric.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

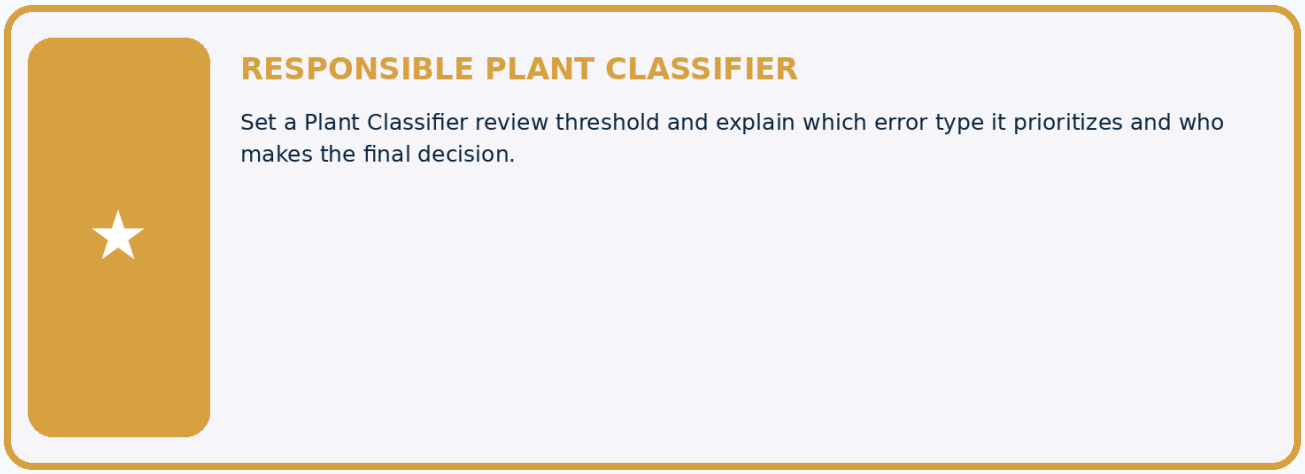
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with the two questions and their counted groups

> Explain when each ratio is unavailable and what evidence would be needed.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Choose the **denominator** that answers the question

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
precision	TP divided by TP plus FP	
recall	TP divided by TP plus FN	
false alarm	false-positive prediction	
miss	false-negative prediction	

Closer-explanation terms

Word	Meaning
Precision	True positives divided by all predicted positives
Recall	True positives divided by all reference positives
Denominator	The group counted in the bottom of a fraction

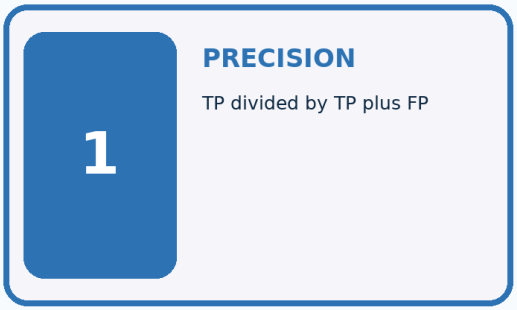
Word	Meaning
Undefined	A ratio with no valid value because its denominator is zero

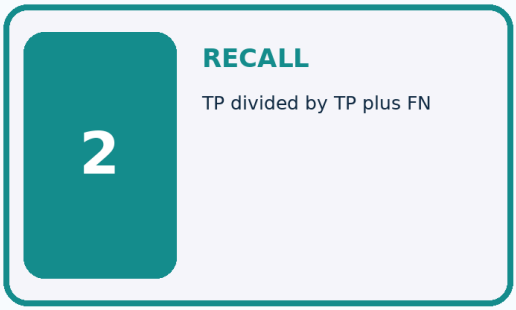
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR **Precision** and **recall** are two names for the same accuracy calculation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **precision**, **recall**, **false alarm**, **miss**.
- CALCULATE Reproduce the worked **evidence** with units and **denominator**.
- TRACE Explain the data, code or decision path.
- DIAGNOSE **Precision** and **recall** are two names for the same accuracy calculation.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

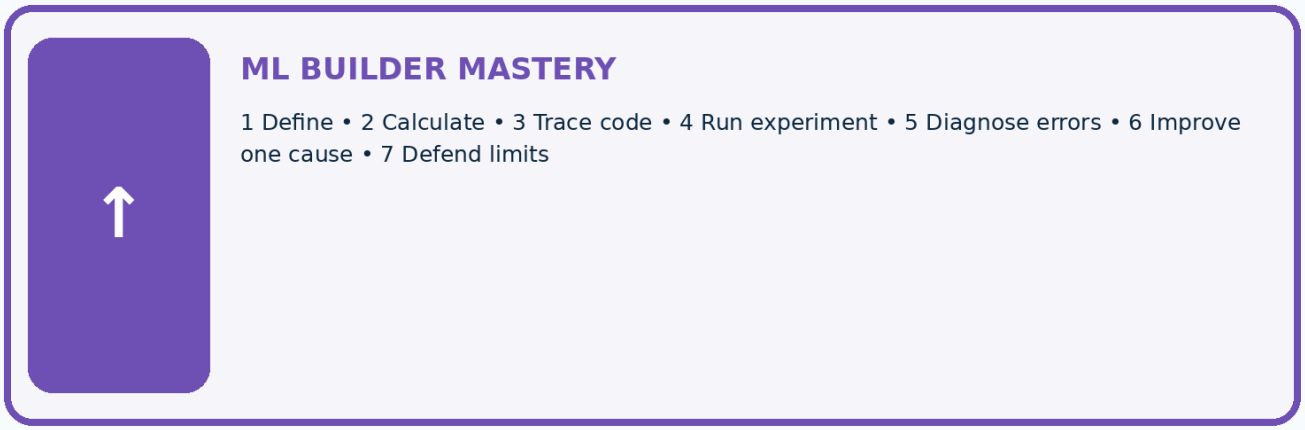
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh19-record precision/recall matrix for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, **miss**ing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

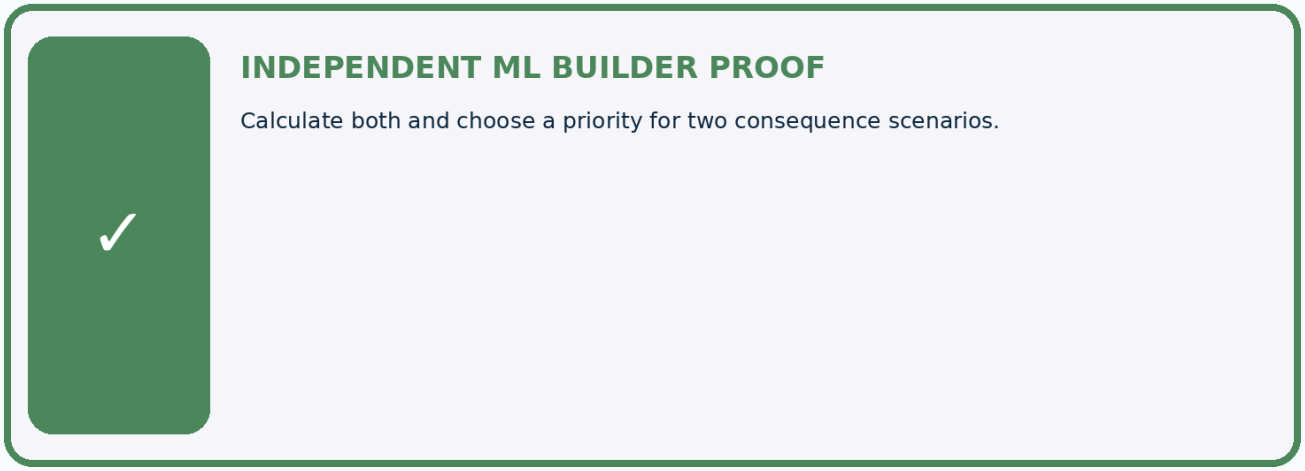
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Calculate both and choose a priority for two consequence scenarios.

NEXT BRIDGE Carry the experiment **evidence** into: Error Analysis.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Positive is spotted: TP=8, FP=2, FN=4.

Compute **precision** and **recall**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

Evidence or stage	Value or result
Precision, predicted positives	8/10
Recall, actual positives	8/12

▼ Earlier wording, alternatives and picture transcripts

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Precision and recall use different denominators

Positive is spotted: TP=8, FP=2, FN=4.

Compute **precision** and **recall**.

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Choose the denominator that answers the question	Explain the deciding step, show a trace or test, and state one limit.
Calculate precision and recall with a declared positive class	Explain the deciding step, show a trace or test, and state one limit.
Report undefined ratios honestly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Put **precision** and **recall denominator**s on the picture instead of memorising names.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Put **precision** and **recall denominator**s on the picture instead of memorising names.

- Which labelled step matches this detail? **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.
- 1. Predicted positives
- 2. Actual positives
- 3. Keep goals separate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep goals separate **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Read the picture in order.

Precision asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Repair and keep the record

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

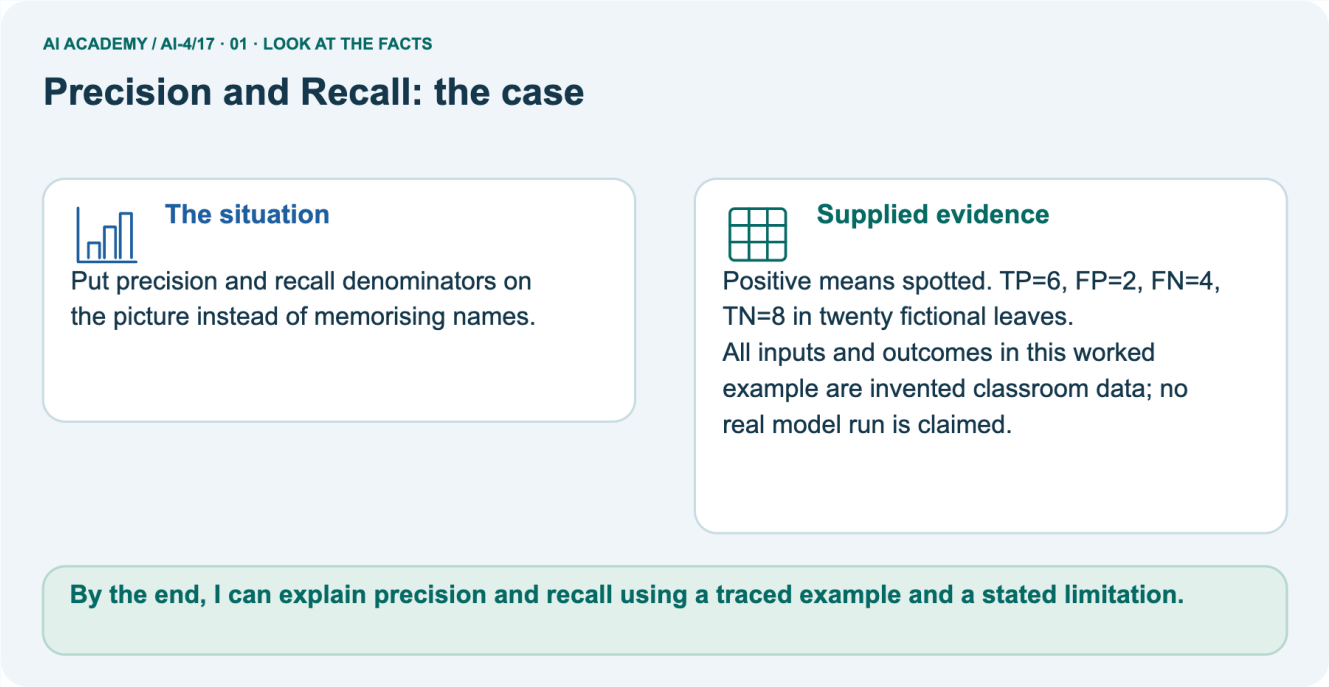
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Precision and Recall: the case



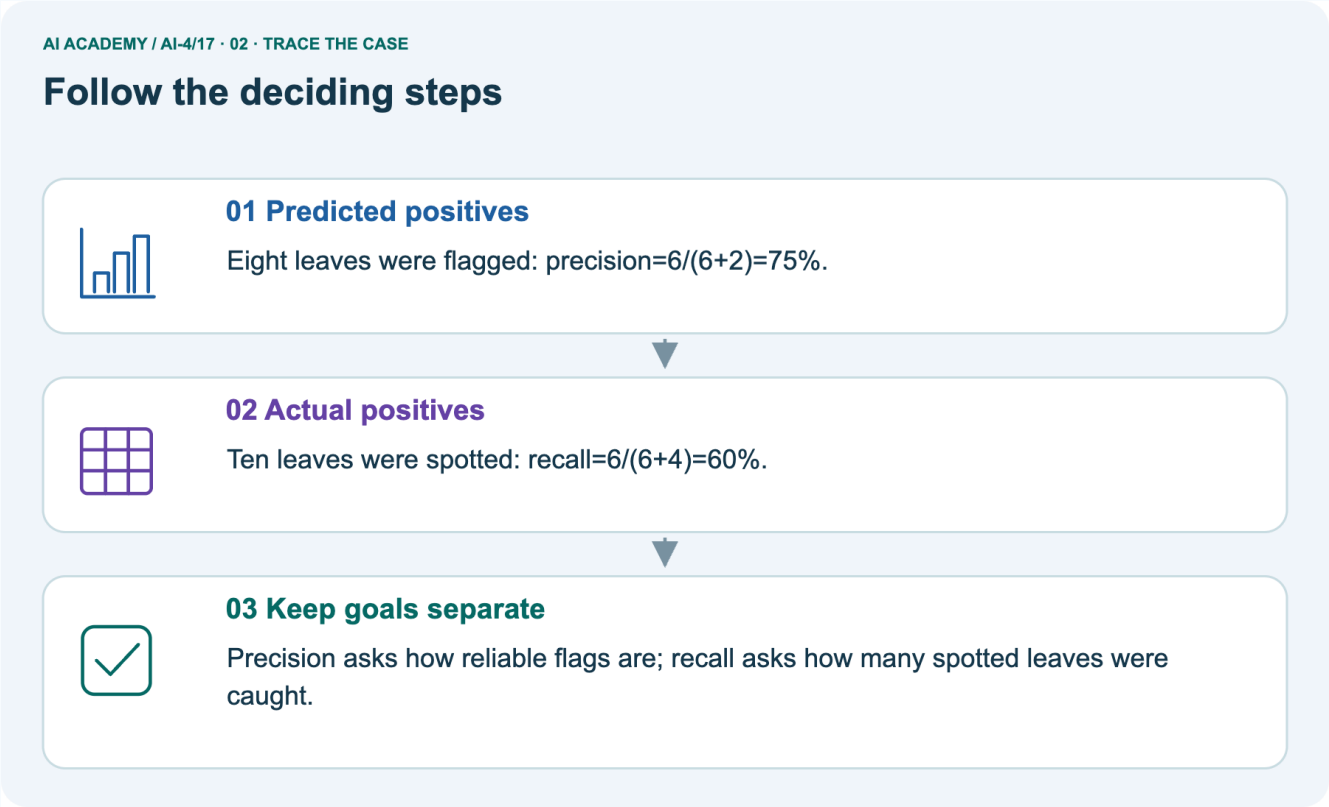
Follow the labelled picture with the matching explanation. Open the vector diagram

Put **precision** and **recall denominator**s on the picture instead of memorising names.
Supplied facts: Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

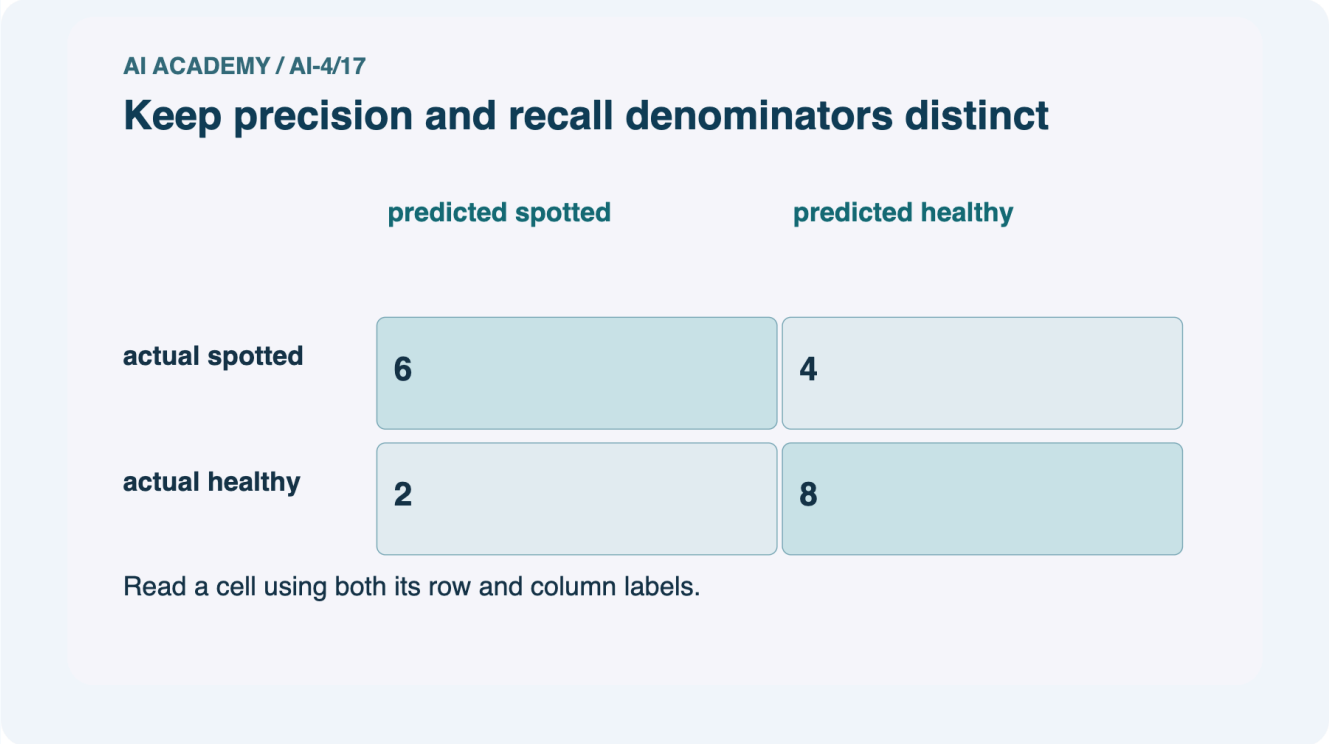


Follow the labelled picture with the matching explanation. Open the vector diagram

Predicted positives: Eight leaves were flagged: **precision**=6/(6+2)=75%. → Actual positives: Ten leaves were spotted: **recall**=6/(6+4)=**60%**. → Keep goals separate: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep precision and recall denominators distinct

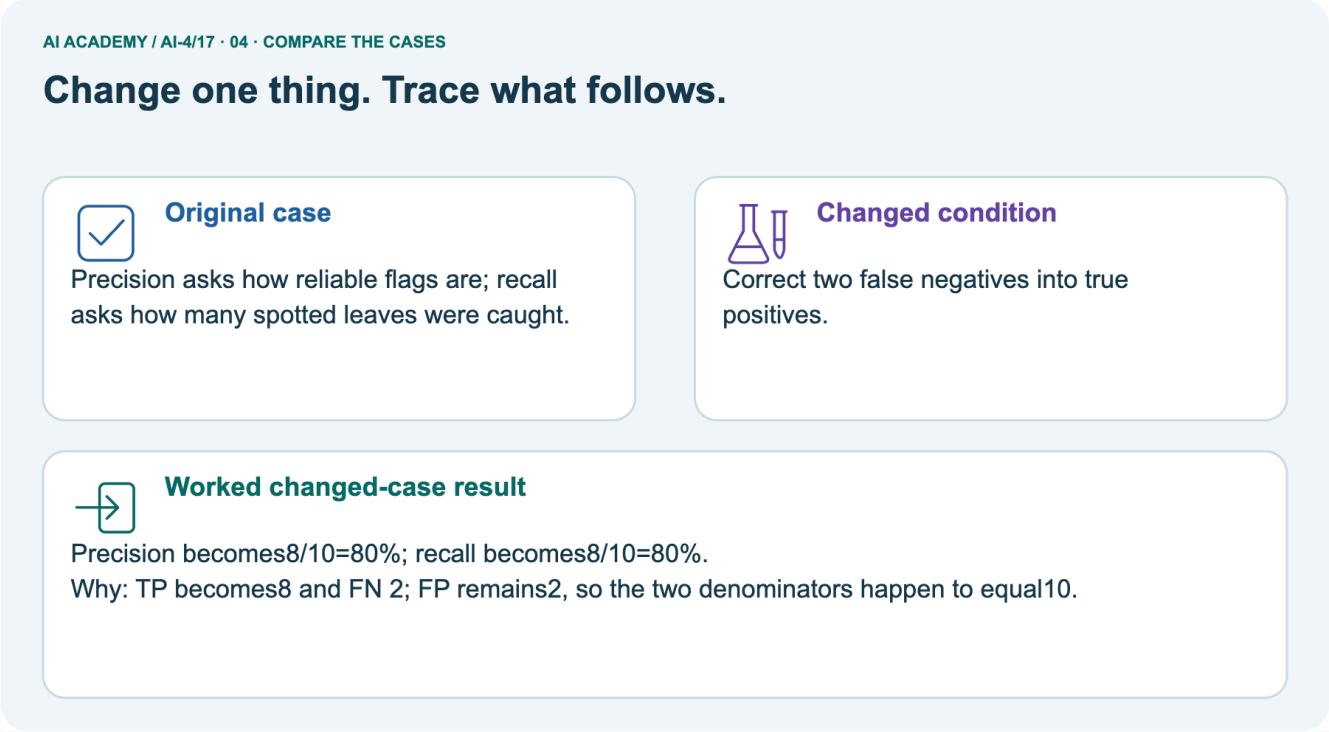


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **precision** and **recall denominator**s distinct. actual spotted: predicted spotted = 6, predicted healthy = 4. actual healthy: predicted spotted = 2, predicted healthy = 8.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



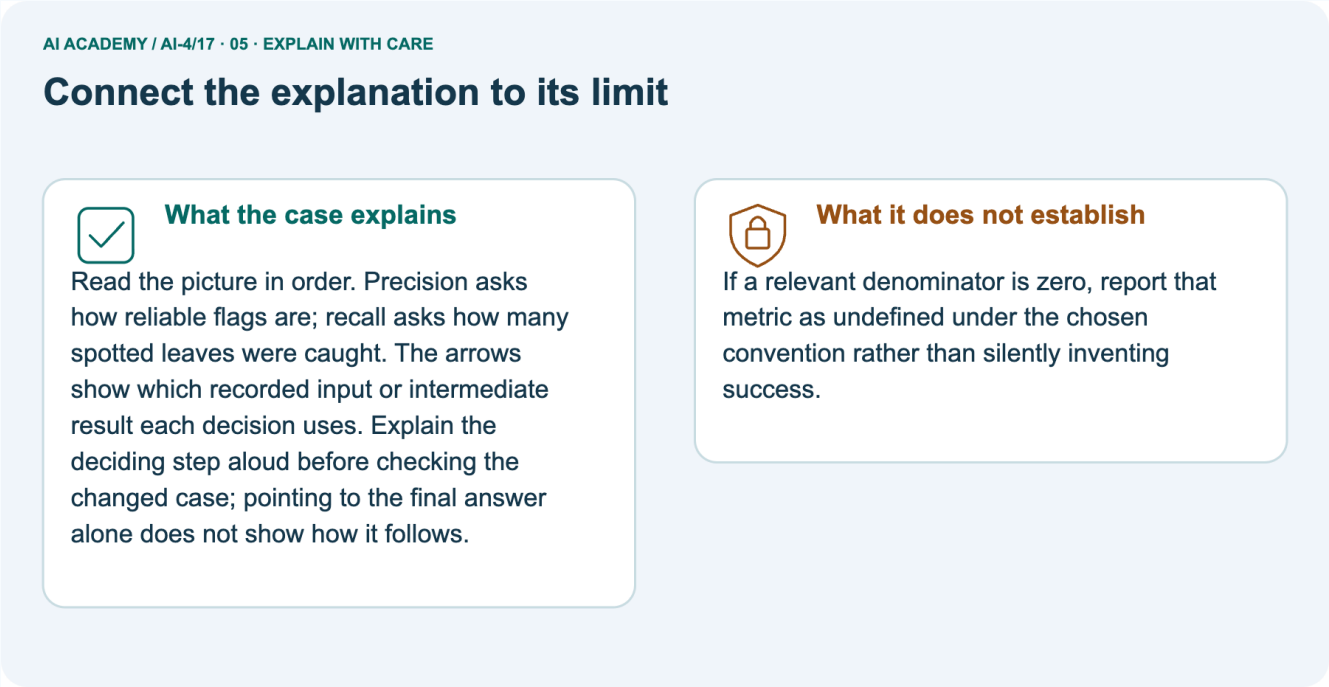
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Changed condition: Correct two false negatives into true positives.

Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

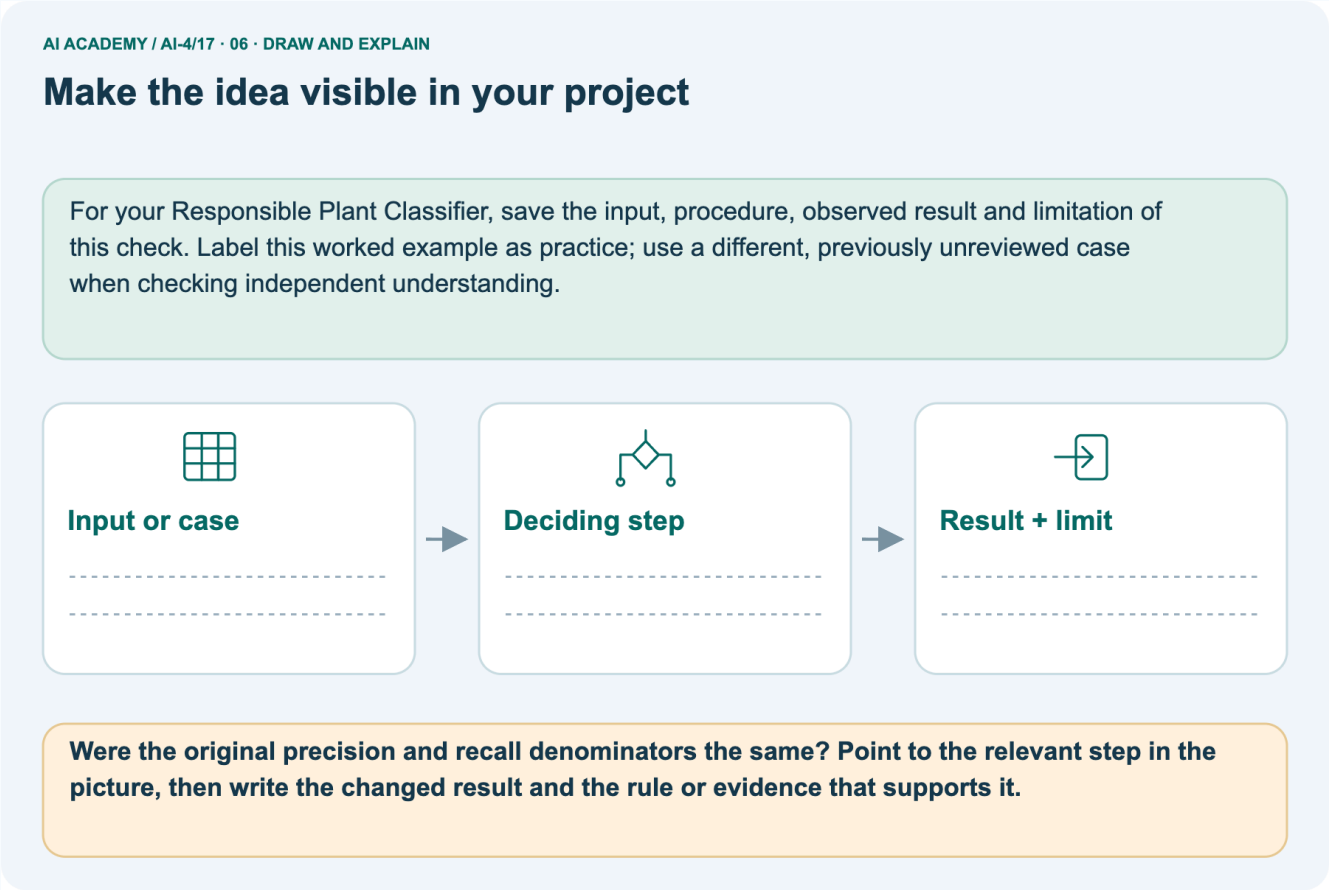
Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Precision and Recall: the case

02. Follow the deciding steps

03. Keep precision and recall denominators distinct

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.